

\$Aborted



```
(** VoS Assignment 2 _ Mathematica Practice Part A by 22ME10083 **)
```

```
(* Practice 1: Defining symbols *)
```

```
(* First checking *)
```

```
Symbols = {L, E, I, rho, A};
```

```
EI = E*I;
```

```
EI
```

```
L
```

```
(* As EI doesnt give any symbolic value  
so I checked if it as inbuilt preassigned value*)
```

```
? E
```

```
? I
```

```
Esym = Symbol["E"]
```

```
Esym
```

```
EI = Symbol["EI"]
```

```
EI
```

```
rho
```

```
 $\rho$  (* 'esc' rho 'esc' gives the sign *)
```

Out[125]=

$i e$

Out[126]=

L

Out[127]=

Symbol 

E is the exponential constant e (base of natural logarithms), with numerical value ≈ 2.71828 .

▼

Out[128]=

Symbol 

I represents the imaginary unit $\sqrt{-1}$.

▼

Out[129]=

e

Out[130]=

e

Out[131]=

$i e$

Out[132]=

$i e$

Out[133]=

ρ

Out[135]=

$$\frac{L^2}{4}$$

(* Practice 1: Defining symbols *)

In[98]:= X; Y; Z

Out[98]=

Z

`ClearAll["Global`*"];`

(* This line will delete all pre allocated values of the varriables *)

`EI = Symbol["EI"];`

EI

Out[138]=

EI

In[1]:= ρ

E

A(* This is done using Esc [Letter] Esc*)

Out[1]= ρ

Out[2]= E

Out[3]= A

In[4]:= `w[x_] := x*(L - x);`

`wAtHalf = w[L/2] // Simplify`

Out[5]= $\frac{L^2}{4}$

(*----- Practice 2: Numerical substitution ----- *)

(* for E and I Esc ... Esc is used*)

$L = 1$; $E = 2 \cdot 10^{11}$; $I = 10^{-6}$;

$EI = E \cdot I$

L

Out[182]=

200 000

Out[183]=

1

(** VoS Assignment 2 _ Mathematica Practice Part B by 22ME10083
**)

(* Practice 3: Creating Vectors and Matrices *)

(* generalized vector*)

$q = \{q_1, q_2, q_3\}$;

q

(*Matrix created*)

$K = \{\{12, -6, 0\}, \{-6, 4, -2\}, \{0, -2, 1\}\}$;

(*Displaying in matrix form*)

MatrixForm[q]

MatrixForm[K]

Out[38]=

$\{q_1, q_2, q_3\}$

Out[40]//MatrixForm=

$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}$

Out[41]//MatrixForm=

$\begin{pmatrix} 12 & -6 & 0 \\ -6 & 4 & -2 \\ 0 & -2 & 1 \end{pmatrix}$

(* Practice 4: Matrix Operations *)

```
M = {{1, 2, 3, 0}, {4, 5, 6, 11}, {7, 8, 9, 12}};
M[[1, 3]]
```

```
Out[43]=
```

```
3
```

```
In[23]:= (*Finding Inverse sum of my last digit 8+3 =11 *)
```

```
A = {{11, 0, 0}, {0, 11, 0}, {0, 0, 11}};
```

```
MatrixForm[A]
```

```
MatrixForm[Inverse[A]]
```

```
MatrixForm[Inverse[A] // N]
```

```
Out[24]//MatrixForm=
```

$$\begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{pmatrix}$$

```
Out[25]//MatrixForm=
```

$$\begin{pmatrix} \frac{1}{11} & 0 & 0 \\ 0 & \frac{1}{11} & 0 \\ 0 & 0 & \frac{1}{11} \end{pmatrix}$$

```
Out[26]//MatrixForm=
```

$$\begin{pmatrix} 0.0909091 & 0. & 0. \\ 0. & 0.0909091 & 0. \\ 0. & 0. & 0.0909091 \end{pmatrix}$$

```
In[44]:= Kq = K.q; (*product of the K and q matrix*)
```

```
MatrixForm[Kq]
```

```
determiatK = Det[K]
```

```
eigenK = Eigenvalues[K]
```

```
Out[45]//MatrixForm=
```

$$\begin{pmatrix} 12 q_1 - 6 q_2 \\ -6 q_1 + 4 q_2 - 2 q_3 \\ -2 q_2 + q_3 \end{pmatrix}$$

```
Out[46]=
```

```
-36
```

```
Out[47]=
```

$$\left\{ \sqrt[3]{15.3...}, \sqrt[3]{2.62...}, \sqrt[3]{-0.898...} \right\}$$

(*Practice 5: Generalized Eigenvalue Problem*)

```
M = DiagonalMatrix[{2, 2, 1}]; (* Mass Matrix*)
(*Eigen value Problem*)
λ = Symbol[];
```

```
K.q = λ*M.q
(* Solve generalized eigenproblem *)
Eigvals = Eigenvalues[{K, M}]
NaturalFreqs = Sqrt[Eigvals];
NaturalFreqs // N
```

 **Set:** Tag Dot in {{12, -6, 0}, {-6, 4, -2}, {0, -2, 1}}.{q1, q2, q3} is Protected.

Out[55]=

```
{2 q1 λ, 2 q2 λ, q3 λ}
```

Out[56]=

```
{ 7.67...,  1.93...,  -0.607...}
```

Out[58]=

```
{2.77028, 1.39011, 0. + 0.77902 i}
```

(* Part C : Repeatative Computations Using A For Loop *)

(* Practice 6: Using a For Loop *)

```

In[63]:= (* Practice 6: Using a For Loop *)
e = E;
F[x_] := x^3 * e^-x;
res = {};
For[x = 0, x ≤ 5, x += 0.5,
f = F[x];
AppendTo[res, {x, f}];
];
TableForm[res, TableHeadings → {None, {"x", "f(x)"}}]

```

Out[67]//TableForm=

x	f(x)
0	0
0.5	0.0758163
1.	0.367879
1.5	0.753064
2.	1.08268
2.5	1.28258
3.	1.34425
3.5	1.29471
4.	1.1722
4.5	1.01231
5.	0.842243

```

In[78]:= (*For Loop to Compute *)
results = {};
For[n = 1, n ≤ 10, n++,

(* Compute f(n) = n^2 / (n + 1) *)
fn = n^2 / (n + 1);

AppendTo[results, {n, fn}];

];
results = results // N
TableForm[results, TableHeadings → {None, {"n", "f(n)"}}]

```

```

Out[80]=
{{1., 0.5}, {2., 1.33333}, {3., 2.25}, {4., 3.2}, {5., 4.16667},
 {6., 5.14286}, {7., 6.125}, {8., 7.11111}, {9., 8.1}, {10., 9.09091}}

```

```

Out[81]//TableForm=


| n   | f(n)    |
|-----|---------|
| 1.  | 0.5     |
| 2.  | 1.33333 |
| 3.  | 2.25    |
| 4.  | 3.2     |
| 5.  | 4.16667 |
| 6.  | 5.14286 |
| 7.  | 6.125   |
| 8.  | 7.11111 |
| 9.  | 8.1     |
| 10. | 9.09091 |


```

■

(* Practice 7:Using Table *)

```

results1 = Table[{x, x^3*Exp[-x]}, {x, 0, 5, 0.5}];
results2 = Table[n^2/(n + 1), {n, 1, 10}];
results1
results2 // N

```

```

Out[90]=
{{0., 0.}, {0.5, 0.0758163}, {1., 0.367879}, {1.5, 0.753064}, {2., 1.08268}, {2.5, 1.28258},
 {3., 1.34425}, {3.5, 1.29471}, {4., 1.1722}, {4.5, 1.01231}, {5., 0.842243}}

```

```

Out[91]=
{0.5, 1.33333, 2.25, 3.2, 4.16667, 5.14286, 6.125, 7.11111, 8.1, 9.09091}

```

Part D : Functtion Defination and Different

(* PRACTICE 8 : *)

```
F[x_] := Sin[x]/(1 + Cos[x]);
dFdx[x_] = D[F[x], x] // Simplify
```

$$\text{Out[4]= } \frac{1}{1 + \cos[x]}$$

```
In[26]:= m = 83;
f[x_] := x ^ m * Exp[-x]
f[x]
(*Computing the derivative*)
dfdx = D[f[x], x] // Simplify
d2fdx2 = D[f[x], {x, 2}] // Simplify
```

$$\text{Out[28]= } e^{-x} x^{83}$$

$$\text{Out[29]= } -e^{-x} (-83 + x) x^{82}$$

$$\text{Out[30]= } e^{-x} x^{81} (6806 - 166 x + x^2)$$

(* Practice 9: Shape Functions *)

```
In[68]:= ϕ[x_] := x ^ 2 * (1 - x);
ϕ1 = D[ϕ[x], x] // Simplify
ϕ2 = D[ϕ[x], {x, 2}] // Simplify
ϕvalue = ϕ[1/2]
ϕ1value = ϕ1 /. x -> 1/2
ϕ2value = ϕ2 /. x -> 1/2
Plot[ϕ[x], {x, 0, 1}, PlotLabel -> {"ϕ vs x"}]
Plot[ϕ1, {x, 0, 1}, PlotLabel -> {"ϕ1 vs x"}]
Plot[ϕ2, {x, 0, 1}, PlotLabel -> {"ϕ2 vs x"}]
```

$$\text{Out[69]= } (2 - 3 x) x$$

Out[70]=

$$2 - 6 x$$

Out[71]=

$$\frac{1}{8}$$

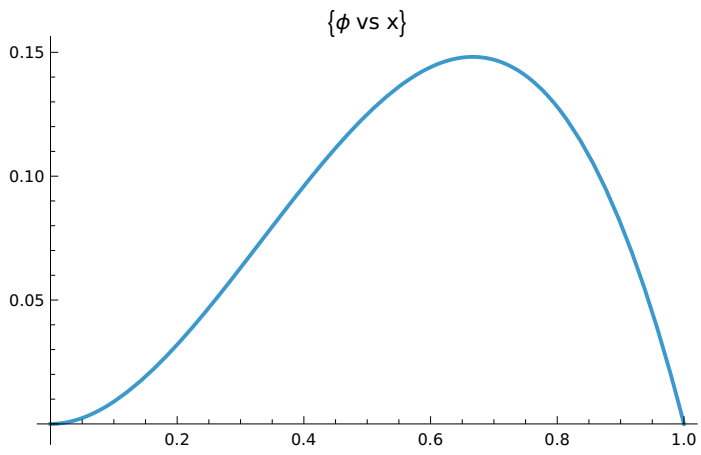
Out[72]=

$$\frac{1}{4}$$

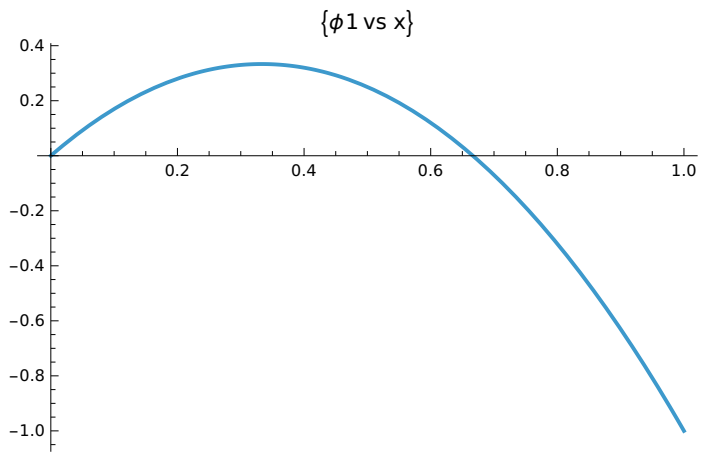
Out[73]=

$$-1$$

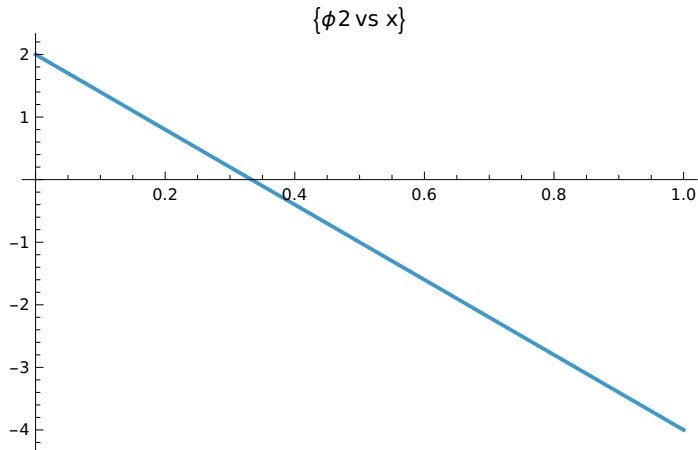
Out[74]=



Out[75]=



Out[76]=



Practice 10: Parametric Functions

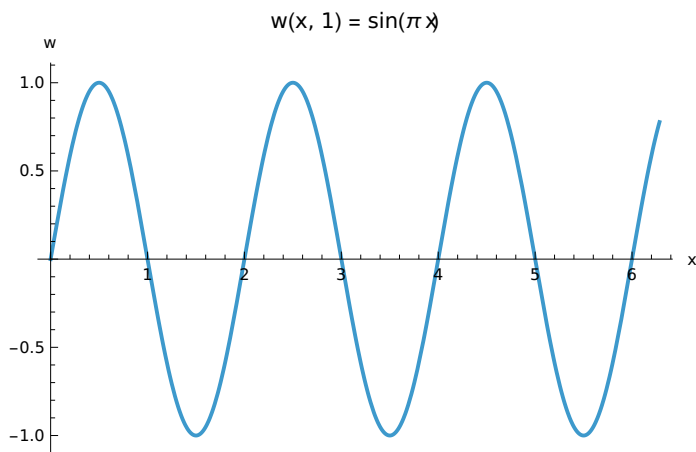
In[100]:=

```
W[x_, a_] := a Sin[Pi * x];
W[x, 1]
Plot[W[x, 1], {x, 0, 2 Pi}, PlotLabel -> "w(x, 1) = sin(pi x)", AxesLabel -> {"x", "w"}]
```

Out[101]=

Sin[πx]

Out[102]=



Part E — Solving Equations

(*Practice 11: Algebraic Equation *)

In[123]:=

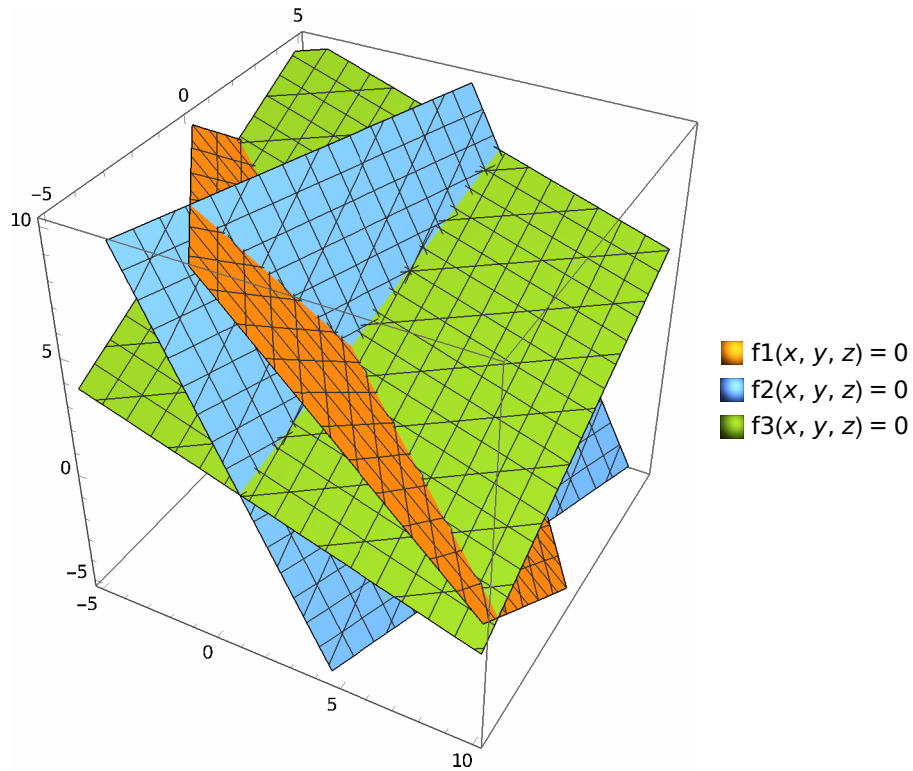
```
Clear[x, y, z];
sol = Solve[{x + y + z == 5, 2 x - y + z == 9, x - 2 y + 3 z == 16}, {x, y, z}]
(* Lets try to graphically understand*)
```

```
f1[x_, y_, z_] := x + y + z - 5;
f2[x_, y_, z_] := 2 x - y + z - 9;
f3[x_, y_, z_] := x - 2 y + 3 z - 16;
ContourPlot3D[{f1[x, y, z] == 0, f2[x, y, z] == 0, f3[x, y, z] == 0},
  {x, -5, 10}, {y, -5, 5}, {z, -5, 10},
  PlotLegends -> "Expressions"]
```

Out[124]=

```
{{x -> 2, y -> -1, z -> 4}}
```

Out[128]=



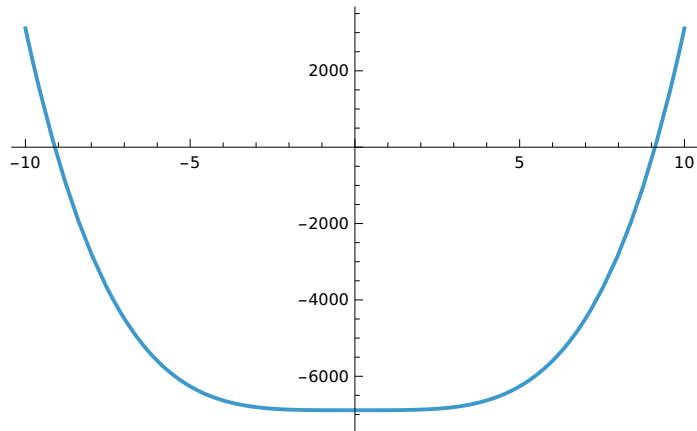
In[132]:=

```
(* Solving for real roots *)
clear[f, x, m];
f[x_] := (x^2 + 83)*(x^2 - 83);
Solve[f[x] == 0, Reals]
Plot[f[x], {x, -10, 10}]
```

Out[134]=

```
{{x -> -sqrt(83)}, {x -> sqrt(83)}}
```

Out[135]=



```
(*Replacement rules,*)
clear[f, x, y];
f[x_, y_] := x^3 + 2 x y + y^3;
f[3, 5]
```

Out[140]=

182

In[149]:=

```
(* Solve the following equation  $3x^2 - 5x + 2 = 0$ *)
clear[f, x, y];
f[x_] := 3 x^2 - 5 x + 2;
Solve[f[x] == 0, x]
NSolve[f[x] == 0, x]
Plot[f[x], {x, 0, 2}]
```

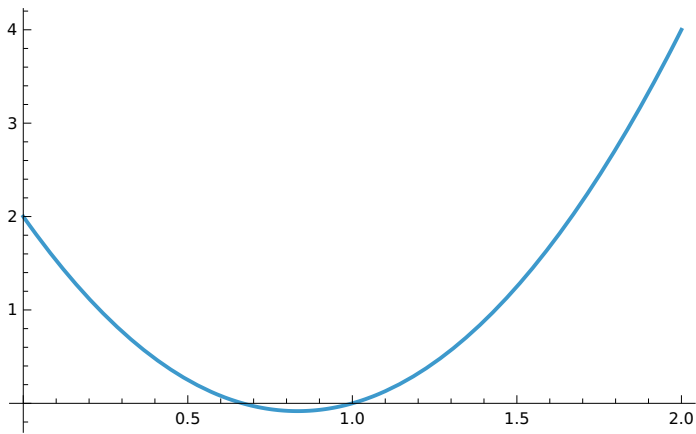
Out[151]=

$$\left\{\left\{x \rightarrow \frac{2}{3}\right\}, \{x \rightarrow 1\}\right\}$$

Out[152]=

$$\{\{x \rightarrow 0.666667\}, \{x \rightarrow 1.\}\}$$

Out[153]=



Practice 12: Transcendental Equation

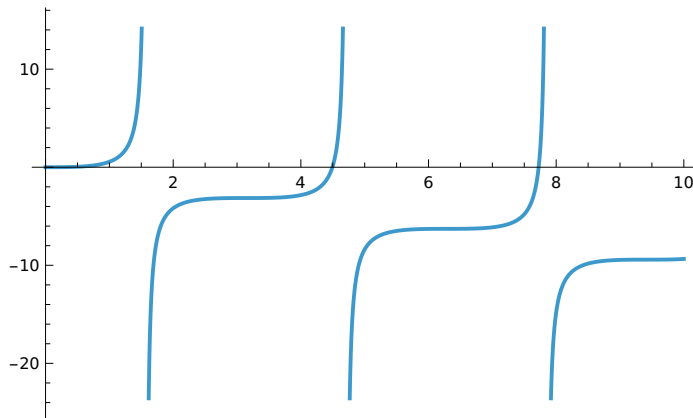
In[210]:=

```

Plot[Tan[x] - x, {x, 0, 10}]
FindRoot[Tan[x] == x, {x, 0.001, 0, 10}]
FindRoot[Tan[x] == x, {x, 4.5, 0, 10}]
FindRoot[Tan[x] == x, {x, 7.7, 0, 10}]
(* Verify from comparing the graph *)
Plot[{Tan[x], x}, {x, 0, 10}, AxesLabel -> {, },
  Epilog -> {Red, PointSize[Large], Point[{1.9752441139975717*^-8, 1.9752441139975717*^-8},
    {4.49341, 4.49341}, {7.72525, 7.72525}]}], PlotLabels -> ]

```

Out[210]=



Out[211]=

 $\{x \rightarrow 1.97524 \times 10^{-8}\}$

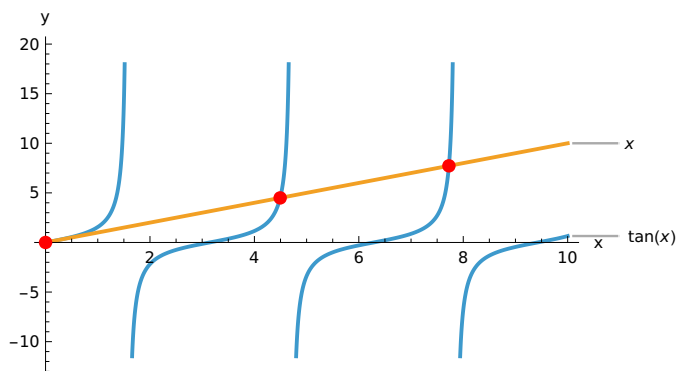
Out[212]=

 $\{x \rightarrow 4.49341\}$

Out[213]=

 $\{x \rightarrow 7.72525\}$

Out[214]=



Part F: Integration and Energy Expressions

(* Practie 13: Indefinite integration *)

```
clear[f, x, y, a];
f[x_] := Exp[2 * x] * Cos[3 * x + 4]
Integrate[f[x], x]
```

Out[219]=

$$\frac{1}{13} e^{2x} (2 \cos[4 + 3x] + 3 \sin[4 + 3x])$$

In[220]:=

```
(* Definite Intergral *)
clear[f, x, y, a];
f[x_] := Sqrt[1 + Sin[2 * x]];
Integrate[f[x], {x, 0, Pi / 2}]
```

Out[222]=

2

(* Practice Problems 14: Potential Energy *)

```

clear[φ, x, y, a];
φ[x_] := x^2*(1 - x);
dφdx = D[φ[x], x] // Simplify
Integrate[dφdx^2, {x, 0, 1}]

(* Now Assume φ is a field variable of springs
or bars then we can write the Potential energy of them as *)
(* For the string case w(x,t)= φ*)
w[x] = φ[x];
dwdx = dφdx;
(* Assuming total length is l and T is not a function of x *)
Vstring = Integrate[0.5*T*dwdx^2, {x, 0, 1}]
(* here actually it is partially derivative*)

(* For the string case u(x,t)= φ *)
(* Assuming total length is l and E,A is not a function of x *)
u[x] = φ[x];
dudx = dφdx;
Vbar = Integrate[0.5*EA*dwdx^2, {x, 0, 1}]

```

Out[33]=

$$(2 - 3x)x$$

Out[34]=

$$\frac{2}{15}$$

Out[37]=

$$0.0666667 T$$

Out[40]=

$$0.0666667 EA$$

(* Practice 15: Kinetic Energy Term *)

```

In[49]:= clear[φ, x, y, a];
φ[x_] := x^2*(1 - x);
dφdx = D[φ[x], x] // Simplify;
Integrate[φ[x]^2, {x, 0, 1}]

```

Out[52]=

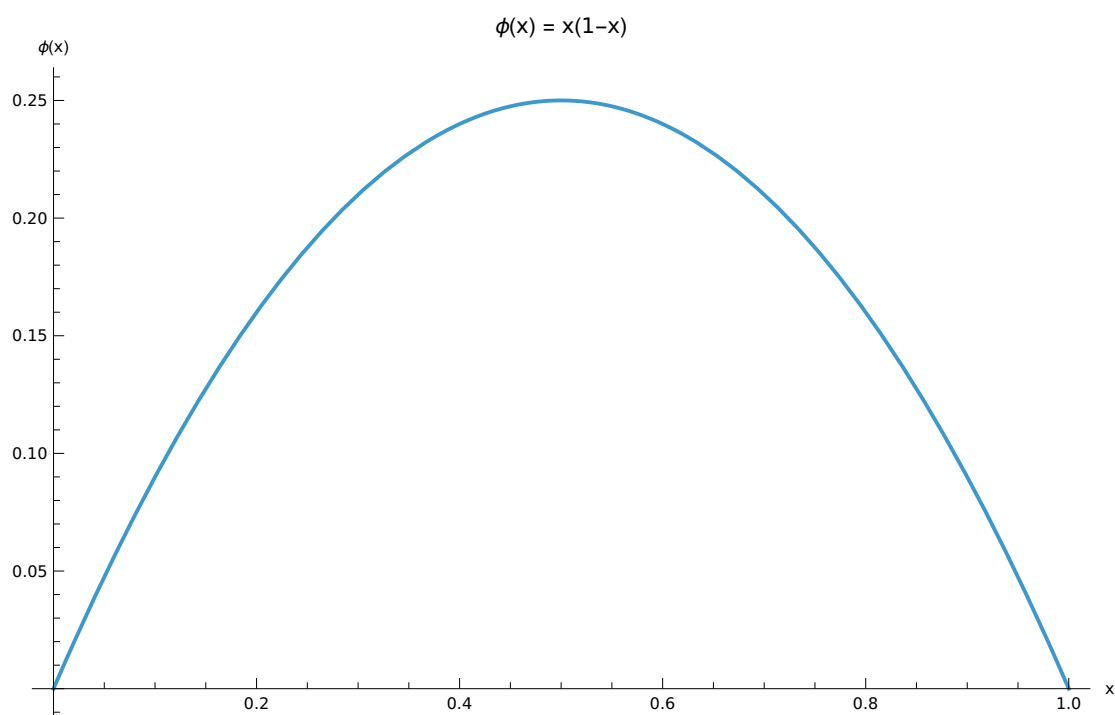
$$\frac{1}{105}$$

Part G: Plotting Mode Shapes

(* Practice 16: Single Function Plot *)

```
(* Single Functional Plot *)  
clear[ $\phi$ , x];  
 $\phi[x_] := x*(1 - x)$ ;  
Plot[ $\phi[x]$ , {x, 0, 1}, PlotRange → All, AxesLabel → {"x", " $\phi(x)$ "},  
PlotLabel → " $\phi(x) = x(1-x)$ ", ImageSize → Large]
```

Out[55]=



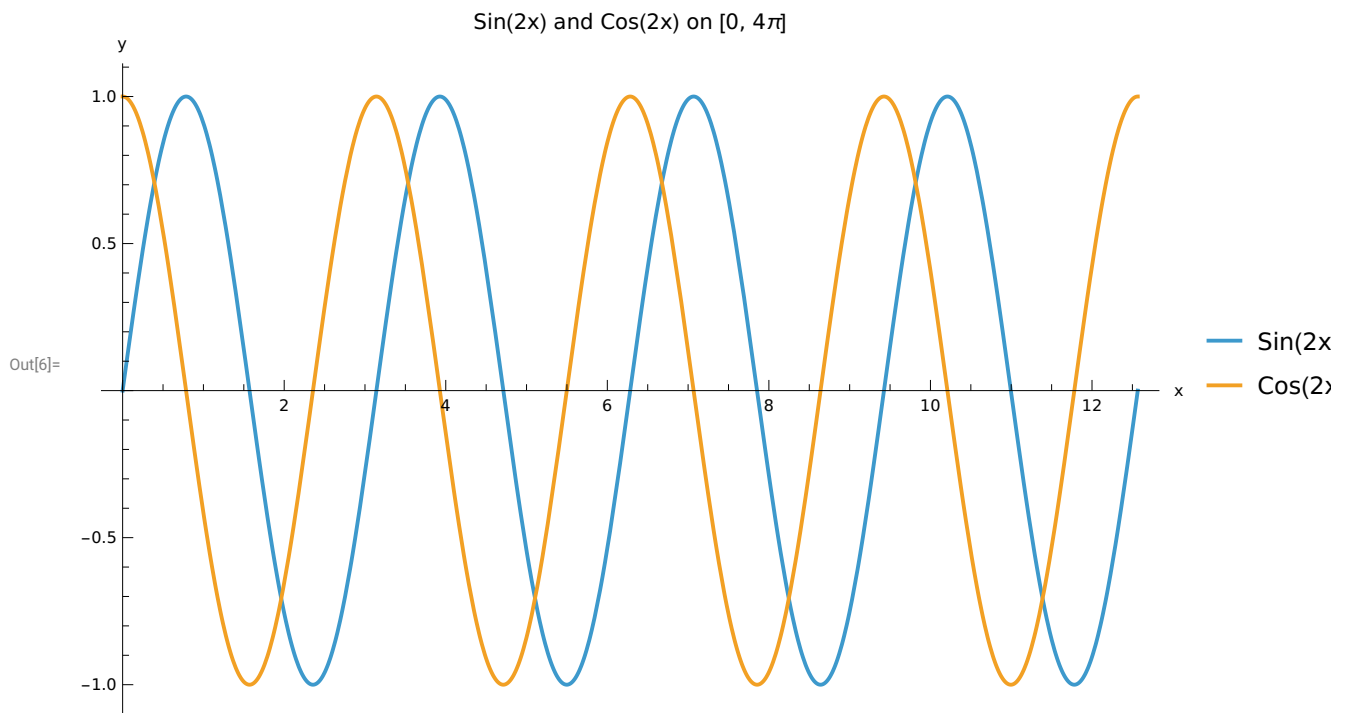
(* Practice 17: Multiple Function Plot *)

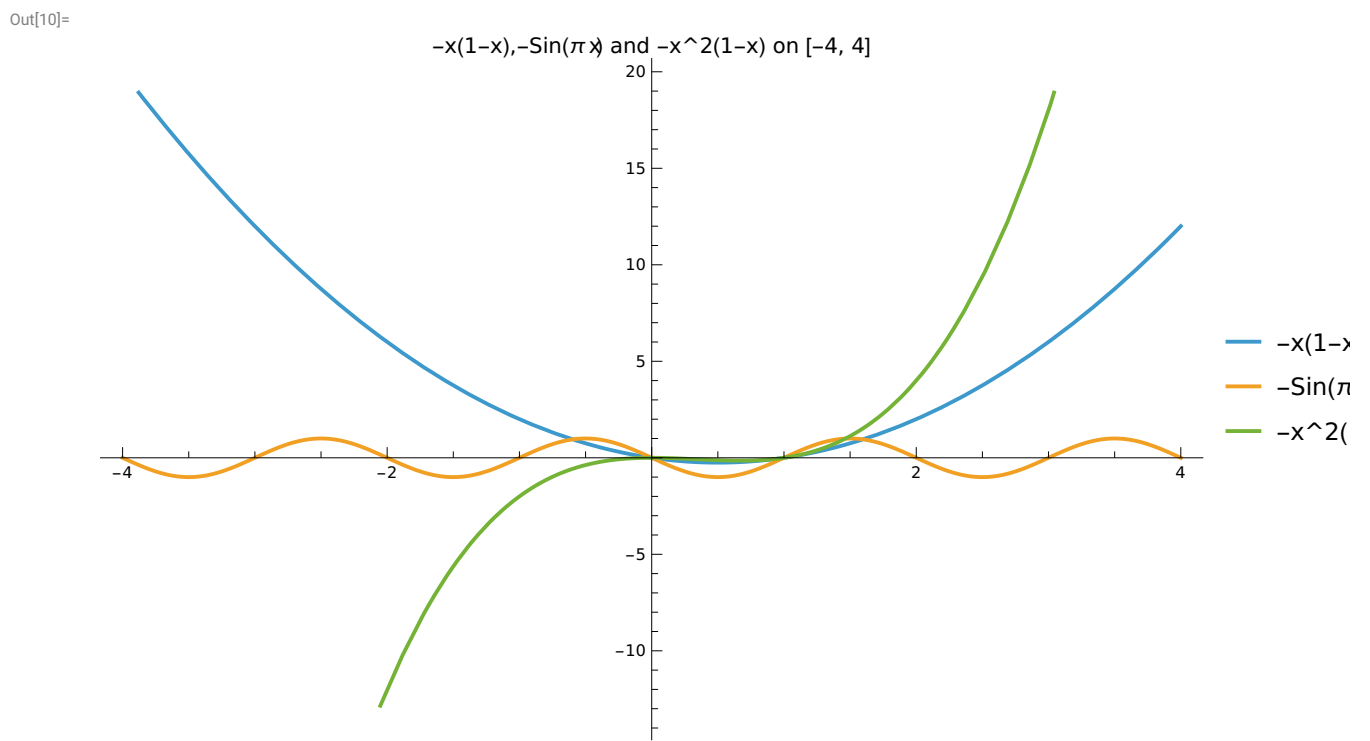
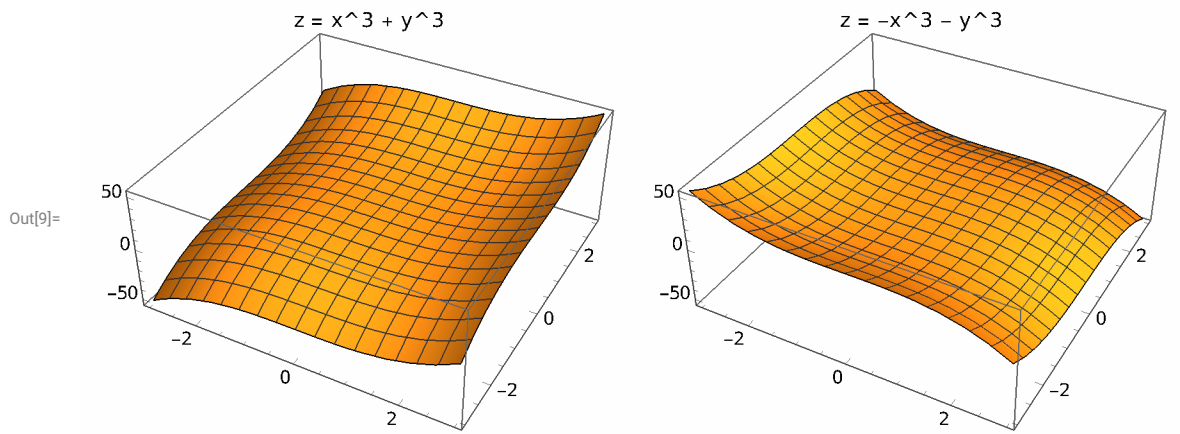
In[6]:=

```
Plot[{Sin[2 x], Cos[2 x]}, {x, 0, 4 Pi}, PlotLegends -> {"Sin(2x)", "Cos(2x)"},
  AxesLabel -> {"x", "y"}, PlotLabel -> "Sin(2x) and Cos(2x) on [0, 4π]", ImageSize -> Large]
```

```
p1 = Plot3D[x^3 + y^3, {x, -3, 3}, {y, -3, 3}, PlotLabel -> "z = x^3 + y^3"];
p2 = Plot3D[-x^3 - y^3, {x, -3, 3}, {y, -3, 3}, PlotLabel -> "z = -x^3 - y^3"];
GraphicsRow[{p1, p2}, ImageSize -> Large]
```

```
Plot[{-x*(1-x), -Sin[Pi*x], -x^2*(1-x)}, {x, -4, 4},
  PlotLegends -> {"-x(1-x)", "-Sin(πx)", "-x^2(1-x)"},
  PlotLabel -> "-x(1-x), -Sin(πx) and -x^2(1-x) on [-4, 4]", ImageSize -> Large]
```





Part H: Ordinary Differential Equations

(* Practice 18: First-Order ODE *)

In[11]:=

```
Clear[x, a];
ySol = DSolve[y'[x] + y[x] == a * Cos[x], y[x], x]
ySol = DSolve[{y'[x] + y[x] == a * Cos[x], y[0] == 0}, y[x], x]

ySol = DSolve[{y'[x] + y[x] == Sin[x], y[0] == 0}, y[x], x]
Plot[y[x] /. ySol, {x, -10, 10}, PlotLabel -> " ODE solved on [-10, 10]", ImageSize -> Large]
```

(* Zoomed plot*)

```
Plot[y[x] /. ySol, {x, -2, 10}, PlotLabel -> , ImageSize -> Large]
```

Out[12]=

$$\left\{ \left\{ y[x] \rightarrow e^{-x} c_1 + \frac{1}{2} a (\cos[x] + \sin[x]) \right\} \right\}$$

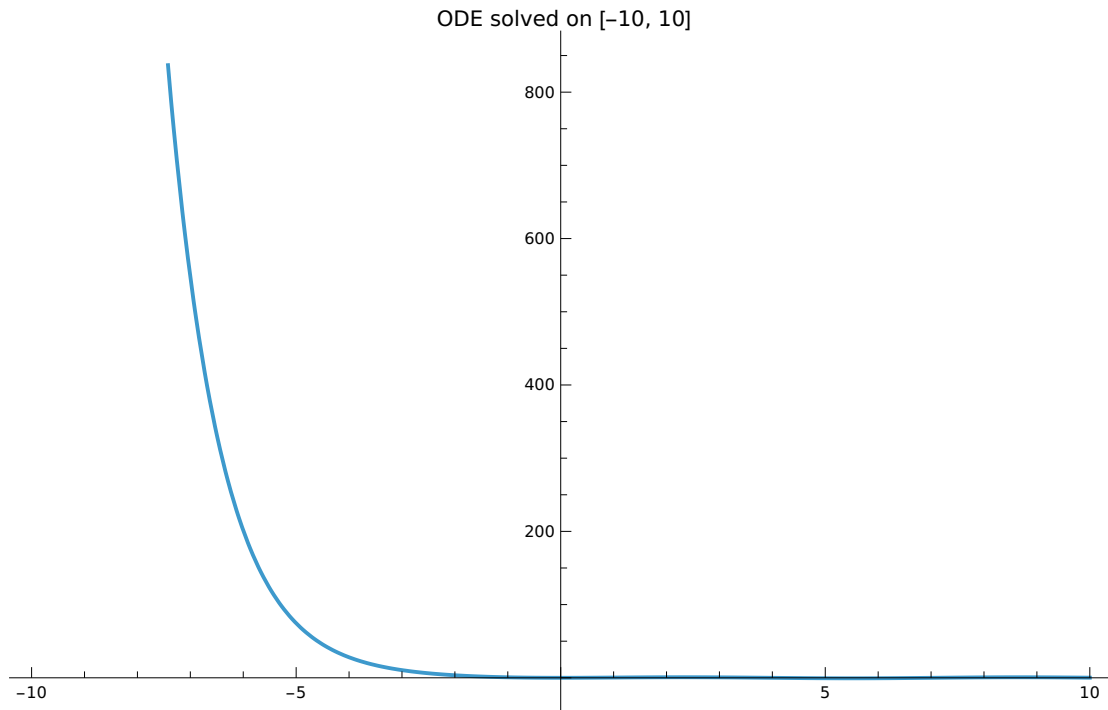
Out[13]=

$$\left\{ \left\{ y[x] \rightarrow \frac{1}{2} a e^{-x} (-1 + e^x \cos[x] + e^x \sin[x]) \right\} \right\}$$

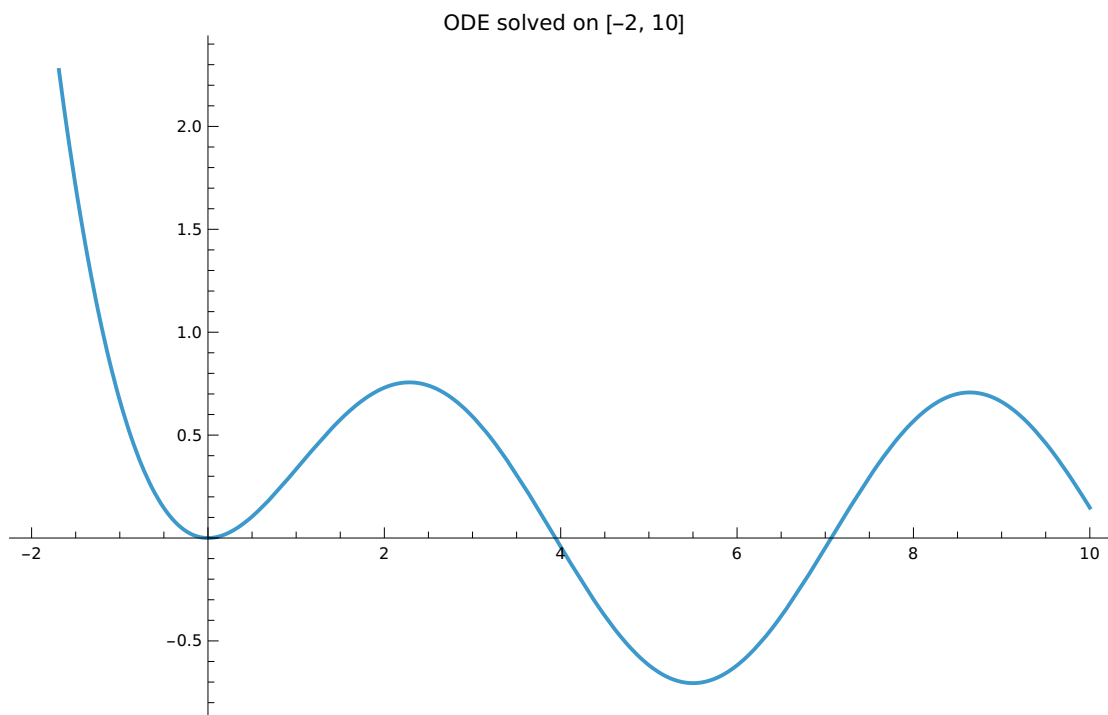
Out[14]=

$$\left\{ \left\{ y[x] \rightarrow -\frac{1}{2} e^{-x} (-1 + e^x \cos[x] - e^x \sin[x]) \right\} \right\}$$

Out[15]=



Out[16]=



(* Practice 19: Second-Order ODE (SDOF System) *)

```

qSol = DSolve[q'[t] + 4 q[t] == 0, q[t], t] // Simplify (* Solve symbolically *)
(* Numerical soln *)
qSol = DSolve[{q'[t] + 4 q[t] == 0, q[0] == 0, q'[0] == 0}, q[t], t] // Simplify

Plot[q[t] /. qSol, {t, 0, 10}, PlotRange → {-0.000000000001, 0.000000000001},
  PlotLabel → " Displacement Vs t on [0, 10]", ImageSize → Large]

```

Out[157]=

```
{{q[t] → c1 Cos[2 t] + c2 Sin[2 t]}}
```

Out[158]=

```
{{q[t] → 0}}
```

Out[159]=

